THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

CIVL2108: PRINCIPLES OF CIVIL ENGINEERING MANAGEMENT

December 12, 2024 6:30 PM-9:30 PM

Answer ALL questions.

A compound interest table is provided.

Use of Calculators:

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

1. a) Explain with the aid of suitable diagram the mechanism of "design and build" and "management contacting". (8 marks)

   b) Please discuss the advantages and disadvantages of the two procurement methods in (a) above. (12 marks)

   c)Discuss the roles and responsibilities of all key involving parties in the stages of Conceptual Planning and Feasibility Study, Design and Engineering, and Procurement. (10 marks)

2. The following table shows the duration and the logical relationship of the activities involved in a construction project:

Activity ID	Predecessor	Duration (weeks)
A	-	5
B	A	10
C	B	20
D	A(SS2), A(FF3)	2
E	D	10
F	C(SS2), E	12
G	F, C(FF3)	4
H	D(SS2), D(FF3)	8
I	H(SS7) H(FF5)	5
J	I, F(FF4)	15
K	J, G	4
L	K	2

In the table above,
SS# = start-to-start relationship with a lag of #
FF# = finish-to-finish relationship with a lag of #

   a) Construct an activity-on-node network diagram based on the information in the table above. (5 marks).
   b) Prepare a table showing ES, EF, LS, and LF of activites D, H, I and F. (16 marks)
   c) Describe what measures should be taken by the project manager to monitor and control construction progress. (4 marks)

3. A scheme has been drawn up for constructing a new toll highway to connect two towns to relieve the increasing and already congested traffic on an existing road connecting the towns. The details of the three options are listed below; please calculate
   a) The present worth of administration cost and replacement cost for the first 10 years. (6 marks)
   b) The present worth of OMR cost for each option for the first 20 years. (3 marks)
   c) The present worth of toll income for the first 20 years. (4 marks)
   d) Find the Option that gives the highest B/C ratio for the first 30 years. (12 marks)

	Option A	Option B	Option C
Capital cost	$650,000,000	$630,000,000	$620,000,000
Annual OMR cost	$3,500,000 per year for the first 10 years, the growth of OMR cost is 5% every 10 years	$4,000,000 per year for the first 10 years, the growth of OMR cost is 5% every 10 years	$6,000,000 per year for the first 10 years, the growth of OMR cost is 5% every 10 years
Annual administration cost	$2,500,000 for the first 30 years		
Replacement cost of autopay ticket machine	$1,000,000 every 5 years		
Estimated annual traffic flow	10,000 vehicles per day in the first five years; the growth rate of traffic flow is 15% every 5 years		
Toll rate	Average $4 for each vehicle		
Life of project	100 years		
Required rate of return (real)	10%		
Estimated inflation rate	10% p.a. (average)		

4. a) Explain the concepts of a matrix organization and project-based/oriented organization. (10 marks)

   b) Modular Integrated Construction (MiC) is an innovative construction method that involves the off-site prefabrication of building modulars or components, which are then transported to the construction site for assembly. Please discuss the advantages of MiC in terms of labour and material management. (10 marks)

Compound Interest Table

$i = 10\%$

n	caf $(1+i)^n$	pwf $\frac{1}{(1+i)^n}$	uscaf $\frac{(1+i)^n - 1}{i}$	ussff $\frac{i}{(1+i)^n - 1}$	uspwf $\frac{(1+i)^n - 1}{i(1+i)^n}$	uscrf $\frac{i(1+i)^n}{(1+i)^n - 1}$
1	1.1000	0.9091	1.0000	1.0000	0.9091	1.1000
2	1.2100	0.8264	2.1000	0.4762	1.7355	0.5762
3	1.3310	0.7513	3.3100	0.3021	2.4869	0.4021
4	1.4641	0.6830	4.6410	0.2155	3.1699	0.3155
5	1.6105	0.6209	6.1051	0.1638	3.7908	0.2638
6	1.7716	0.5645	7.7156	0.1296	4.3553	0.2296
7	1.9487	0.5132	9.4872	0.1054	4.8684	0.2054
8	2.1436	0.4665	11.436	0.0874	5.3349	0.1874
9	2.3579	0.4241	13.579	0.0736	5.7590	0.1736
10	2.5937	0.3855	15.937	0.0627	6.1446	0.1627
11	2.8531	0.3505	18.531	0.0540	6.4951	0.1540
12	3.1384	0.3186	21.384	0.0468	6.8137	0.1468
13	3.4523	0.2897	24.523	0.0408	7.1034	0.1408
14	3.7975	0.2633	27.975	0.0357	7.3667	0.1357
15	4.1772	0.2394	31.772	0.0315	7.6061	0.1315
16	4.5950	0.2176	35.950	0.0278	7.8237	0.1278
17	5.0545	0.1978	40.545	0.0247	8.0216	0.1247
18	5.5599	0.1799	45.599	0.0219	8.2014	0.1219
19	6.1159	0.1635	51.159	0.0195	8.3649	0.1195
20	6.7275	0.1486	57.275	0.0175	8.5136	0.1175
21	7.4002	0.1351	64.002	0.0156	8.6487	0.1156
22	8.1403	0.1228	71.403	0.0140	8.7715	0.1140
23	8.9543	0.1117	79.543	0.0126	8.8832	0.1126
24	9.8497	0.1015	88.497	0.0113	8.9847	0.1113
25	10.835	0.0923	98.347	0.0102	9.0770	0.1102
26	11.918	0.0839	109.18	0.0092	9.1609	0.1092
27	13.110	0.0763	121.10	0.0083	9.2372	0.1083
28	14.421	0.0693	134.21	0.0075	9.3066	0.1075
29	15.863	0.0630	148.63	0.0067	9.3696	0.1067
30	17.449	0.0573	164.49	0.0061	9.4269	0.1061

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